

Interpretable and Efficient Penalized Multivariate Monomial Splines

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Abstract

We propose a computationally efficient method for fitting penalized piecewise polynomial splines directly in monomial form, where smoothness at partition boundaries is enforced through explicit linear constraints on the coefficients rather than construction of a smooth basis. The constrained penalized estimate is obtained via a transformed ordinary least squares projection that exploits block-diagonal structure across partitions, avoiding large dense systems and enabling parallel computation. A closed-form integrated curvature penalty on the monomial basis is used for regularization, and the block-diagonal structure is preserved for generalized linear models, inequality constraints, and certain covariance structures. In a simulation study, the proposed method is competitive with tensor product cubic regression and thin-plate spline constructions in terms of performance with improvements in runtimes at larger sample sizes. In a pharmacokinetic application, the monomial framework supports simultaneous shape constraints, structured within-subject correlation, and direct extraction of non-compartmental parameter summaries with fully propagated posterior uncertainty in a single fit. Implementation is available in the `lgspline` package for R.

Keywords: Constrained optimization; Vandermonde basis; Lagrangian multipliers; Block-diagonal estimation; Shape-constrained splines; Smoothing spline penalty. Woodbury identity.

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1 Introduction

Splines are widely used for modeling nonlinear predictor-response relationships across statistics and applied science. Although piecewise polynomial fitting dates to the 19th century (Farin, 2002), modern spline methodology is largely built around basis constructions that enforce smoothness implicitly, most notably B-splines (Cox, 1972; de Boor, 1972). These representations are numerically stable and computationally efficient, and are the foundation of widely used software such as `mgcv` (Wood, 2017) and `splines` (R Core Team, 2025). Truncated power bases span the same function space and are more straightforward to implement without specialized software for B-spline bases, but are less computationally stable (Stone and Koo, 1985; Ezhov, Neitzel, and Petrovic, 2021). Related approaches for fitting splines include M- and I-splines (Schumaker, 2007; Ruppert, Wand, and Carroll, 2003; Ramsay, 1988), thin-plate splines (Duchon, 1977; Wahba, 1990), and smoothing spline ANOVA (Gu, 2002), among many others. Chapters 4 and 5 of Wood (2017) may be referenced for an introductory history and overview of spline construction in the context of generalized additive models (GAMs), and Ezhov, Neitzel, and Petrovic (2021) provides the explicit relationship between monomial, truncated power, and B-spline bases.

For many spline constructions, including B-splines and truncated power splines, the fitted function is a sum of piecewise polynomials and can in principle be rearranged into a monomial basis (Ezhov, Neitzel, and Petrovic, 2021). In standard implementations, however, this structure is implicit and not always accessible. Recovering explicit polynomial expressions requires nontrivial post-fitting algebraic rearrangement that is generally unavailable in modern software. This difficulty becomes substantial in multivariate settings with interactions, where tensor product bases lead to combinatorial growth in the complexity of the bookkeeping, with additional work needed to calculate standard errors. For other spline constructions such as thin-plate splines, rearrangement of the fitted function into polynomial form is not possible at all.

Expressing the fitted model directly in monomial form, sometimes referred to as Vandermonde or power form, provides an immediately interpretable representation with natural extensions to higher dimensions through simple interaction terms. In this form, derivatives,

integrals, and extrema are straightforward to express, and the model can be communicated or evaluated without reference to knots, basis functions, or the method used to fit it.

In contrast, B-splines and truncated power splines are often considered difficult to interpret directly from the model output in terms of basis expansions and coefficients alone (Ezhov, Neitzel, and Petrovic, 2018; Acharjee and Das, 2022). While many statisticians are comfortable viewing splines through a purely nonparametric lens, a parametric interpretation is often desired in practice by subject-matter researchers. Examples include applications to pharmacokinetic modeling (Yeh and Small, 1989; Popović, 2006), epidemiology (Schuster, Rijnhart, Twisk, and Heymans, 2022), and exposure-response analyses (Malloy, Kapellusch, and Garg, 2017). For clinical trial submissions in particular, having an interpretable estimand expressible as a linear combination of one or several coefficients from a parametric model targeting a predictor-response relationship, and that can be written out cleanly with notation for a statistical analysis plan, is not optional for many regulatory bodies, which B-splines and truncated power bases are not ideal for.

An alternative formulation of splines as piecewise polynomial regression models subject to linear restrictions on the coefficients enforcing smoothness was explored early on by Buse and Lim (1977) and Fuller (1969). Ezhov, Neitzel, and Petrovic (2018) showed that the linear restrictions can be expressed as external constraints and the polynomial regression spline can be fit directly in monomial form using the method of Lagrangian multipliers. They do not take credit for the method, suggesting it was so intuitive that it had likely been rediscovered several times over, but they could not find a previous reference. A similar constrained monomial formulation was independently developed by the author (Davis, 2023) as a component of a hyperparameter optimization algorithm, but was not the focus of the work. The approach has otherwise seen limited adoption, perhaps because the spline basis in the single-predictor case is a poorly conditioned Vandermonde matrix, the approach requires the additional step of constructing external constraints, and the resulting system of equations is much greater in dimension than for B-splines or truncated power splines when fit naively. To the best of our knowledge, the monomial formulation has never been formally explored with an arbitrary numbers of predictors, penalization, nor outside of the

scenario of Gaussian-distributed error terms.

In this article we develop a computational framework that revisits the constrained formulation and makes direct fitting in monomial form computationally practical. The key idea is to exploit a partition-wise block-diagonal structure and compute the constrained estimate via a transformed ordinary least squares projection. This avoids construction of large dense systems and enables parallel computation across partitions for most model-fitting operations. The procedure extends straightforwardly to generalized linear models, shape constraints, and the incorporation of certain covariance structures.

The specific contributions are as follows. First, we develop a constrained penalized estimation framework in which smoothness is enforced through explicit linear constraints. Second, we derive a transformed OLS projection that exploits block-diagonal structure for computational and memory efficiency. Third, we establish extensions that preserve this structure under GLM weights, partition-local inequality constraints, and structured covariance via a Woodbury decomposition. Fourth, we provide a closed-form curvature penalty defined on the monomial basis. Fifth, we present empirical studies demonstrating predictive performance, computational scaling, and practical interpretability.

The remainder of the paper is organized as follows. Section 2 describes the model formulation. Section 3 presents the computational strategy, extensions for generalized linear models and structured covariance, and Bayesian inference under constraint. Section 4 discusses practical considerations. Section 5 presents empirical results, beginning with a simulation study, followed by a pharmacokinetic application, a parallelism benchmark, and a basis conditioning analysis. Section 6 concludes.

2 Model Formulation

2.1 Piecewise Polynomial Structure

Consider data consisting of N observations with q -dimensional predictor vectors $\mathbf{t}_i \in \mathbb{R}^q$ and scalar responses $y_i \in \mathbb{R}$, let f be some unknown smooth function $f : \mathbb{R}^q \rightarrow \mathbb{R}$, and suppose that $y_i \mid \mathbf{t}_i, \sigma^2 \sim \mathcal{N}(f(\mathbf{t}_i), \sigma^2)$ with $\text{Cov}(y_i, y_{i'}) = 0$ for $i \neq i'$. We wish to estimate

f .

First, we partition the predictor space into $K + 1$ mutually exclusive regions $\mathcal{P}_0, \dots, \mathcal{P}_K$, so that each observation belongs to exactly one partition. Within partition k , we approximate f with a polynomial:

$$\hat{f}_k(\mathbf{t}) = \mathbf{x}(\mathbf{t})^\top \boldsymbol{\beta}_k, \quad (1)$$

where $\mathbf{x}(\mathbf{t}) \in \mathbb{R}^p$ collects monomial basis terms (intercept, polynomial, and interaction terms as needed) and $\boldsymbol{\beta}_k \in \mathbb{R}^p$ is the coefficient vector for partition k . The $i = 1, \dots, n_k$ expansions $\mathbf{x}_i$ in partition k can be expressed as an $n_k \times p$ design matrix $\mathbf{X}_k = (\dots, \mathbf{x}_i, \dots)^\top$.

In the simple case of a single predictor and one partition with cubic expansions, $\mathbf{x}(t) = (1, t, t^2, t^3)^\top$ and the fitted function is $\hat{f}(t) = \hat{\beta}_0 + \hat{\beta}_1 t + \hat{\beta}_2 t^2 + \hat{\beta}_3 t^3$. Derivatives follow directly from the power rule, for example $\hat{f}'(t) = \mathbf{x}'(t)^\top \hat{\boldsymbol{\beta}} = \hat{\beta}_1 + 2\hat{\beta}_2 t + 3\hat{\beta}_3 t^2$. Unlike B-splines and truncated powers, for this formulation the expansion $\mathbf{x}(\mathbf{t})$ is the same in every partition.

Let the full coefficient vector be $\boldsymbol{\beta} = (\boldsymbol{\beta}_0^\top, \dots, \boldsymbol{\beta}_K^\top)^\top \in \mathbb{R}^P$ with $P = p(K+1)$. The $N \times P$ design matrix is block-diagonal, $\mathbf{X} = \text{blockdiag}(\mathbf{X}_0, \dots, \mathbf{X}_K)$, with blocks of dimension $n_k \times p$ where total sample size $N = \sum_{k=0}^K n_k$.

Let $\boldsymbol{\Lambda} = \text{blockdiag}(\boldsymbol{\Lambda}_0, \dots, \boldsymbol{\Lambda}_K)$ be a block-diagonal penalty matrix of dimension $P \times P$ with positive-definite $p \times p$ blocks (see 2.3). The model can be fit using penalized least squares:

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta}} \left\{ \frac{1}{2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \frac{1}{2} \boldsymbol{\beta}^\top \boldsymbol{\Lambda} \boldsymbol{\beta} \right\}. \quad (2)$$

The solution is the usual penalized maximum likelihood (MLE) estimate, equivalently, the maximum *a posteriori* estimate, $\hat{\boldsymbol{\beta}} = \mathbf{G}\mathbf{X}^\top \mathbf{y}$, where $\mathbf{G} = (\mathbf{X}^\top \mathbf{X} + \boldsymbol{\Lambda})^{-1}$ is block-diagonal with positive-definite $p \times p$ blocks $\mathbf{G}_k = (\mathbf{X}_k^\top \mathbf{X}_k + \boldsymbol{\Lambda}_k)^{-1}$. For each partition $\mathcal{P}_k$ the penalized estimate is simply $\hat{\boldsymbol{\beta}}_k = \mathbf{G}_k \mathbf{X}_k^\top \mathbf{y}_k$, so fitting can proceed in parallel across partitions.

2.2 Smoothness Constraints

Without constraint the piecewise polynomial is discontinuous at partition boundaries. Smoothness is imposed by requiring the polynomial and its derivatives to agree at shared

knot locations (Fuller, 1969; Buse and Lim, 1977; Ezhov, Neitzel, and Petrovic, 2018). At each knot ξ between neighboring partitions k and k^* , the function and its first $d - 1$ derivatives are constrained to match:

$$f_k(\xi) = f_{k^*}(\xi), \quad f'_k(\xi) = f'_{k^*}(\xi), \quad \dots, \quad f_k^{(d-1)}(\xi) = f_{k^*}^{(d-1)}(\xi). \quad (3)$$

For example, for a cubic spline with $d = 3$, this enforces continuity of the function, its first derivative, and its second derivative. With multiple predictors, **lgspline** applies predictor-specific first-derivative matching and a total second-derivative matching rule in which pure and mixed second-order terms contributing to a given predictor are combined. Thus the number of constraints grows with the number of predictors and neighboring partitions, but not with the number of polynomial terms nor interaction terms included in the expansions.

These constraints are linear in $\boldsymbol{\beta}$ (Fuller, 1969; Buse and Lim, 1977; Ezhov, Neitzel, and Petrovic, 2018) and can be collected into a matrix $\mathbf{A}$ of dimension $P \times R$ such that $\mathbf{A}^\top \boldsymbol{\beta} = \mathbf{0}$ implies the smoothness conditions are satisfied, illustrated in Supplement A. A pivoted QR decomposition of $\mathbf{A}$ improves computational stability and prevents redundant constraints in practice. The constraint matrix is typically underdetermined ($R < P$) for practical applications, with the model dimension often growing faster in q and K than the constraint count (see Supplement A.2 for details). Columns can be added to $\mathbf{A}$ for user-supplied equality constraints or removed to relax specific smoothness conditions, a flexibility that is unique and central to the framework.

The constrained solution follows from Lagrangian multipliers (Appendix A). Define the projection

$$\mathbf{U} = \mathbf{I} - \mathbf{GA}(\mathbf{A}^\top \mathbf{GA})^{-1} \mathbf{A}^\top, \quad (4)$$

where $\mathbf{G} = \text{blockdiag}(\mathbf{G}_0, \dots, \mathbf{G}_K)$. The constrained estimate is

$$\tilde{\boldsymbol{\beta}} = \mathbf{U} \hat{\boldsymbol{\beta}}. \quad (5)$$

Fitted values are $\tilde{\mathbf{y}} = \mathbf{X} \tilde{\boldsymbol{\beta}} = \mathbf{H} \mathbf{y}$ for hat matrix $\mathbf{H} = \mathbf{X} \mathbf{U} \mathbf{G} \mathbf{X}^\top$. For non-zero constraints ($\mathbf{A}^\top \boldsymbol{\beta} = \mathbf{A}^\top \boldsymbol{\beta}_0 = \mathbf{c}$), the solution becomes $\tilde{\boldsymbol{\beta}} = \mathbf{U} \hat{\boldsymbol{\beta}} + (\mathbf{I} - \mathbf{U}) \boldsymbol{\beta}_0$.

The matrix $\mathbf{U}$ is idempotent and satisfies $\mathbf{A}^\top \mathbf{U} = \mathbf{0}$ and $\mathbf{UGU}^\top = \mathbf{UG}$. Effective degrees of freedom are $\nu = \text{tr}(\mathbf{H})$ and for Gaussian identity-link models a variance estimate

is $\tilde{\sigma}^2 = \|\mathbf{y} - \tilde{\mathbf{y}}\|^2 / (N - \nu)$ (Wood, 2017, §6.2.1) can be obtained by maximizing the REML objective as described in Supplement D.

For Gaussian identity-link models, the exact frequentist variance-covariance matrix of $\tilde{\boldsymbol{\beta}}$ is

$$\widetilde{\text{Var}(\tilde{\boldsymbol{\beta}})} = \tilde{\sigma}^2 \mathbf{U} \mathbf{G} (\mathbf{X}^\top \mathbf{X}) \mathbf{G} \mathbf{U}^\top = \tilde{\sigma}^2 \mathbf{U} \mathbf{G} - \tilde{\sigma}^2 \mathbf{U} \mathbf{G} \boldsymbol{\Lambda} \mathbf{G} \mathbf{U}^\top, \quad (6)$$

using $\mathbf{G}(\mathbf{X}^\top \mathbf{X}) \mathbf{G} = \mathbf{G}(\mathbf{G}^{-1} - \boldsymbol{\Lambda}) \mathbf{G}$. The first term $\tilde{\sigma}^2 \mathbf{U} \mathbf{G}$ is the Bayesian posterior variance-covariance matrix under the interpretation of the penalty matrix as a precision matrix for normal priors on coefficients (Wood, 2017) (see 3.3), and subtracting the second term yields the frequentist exact estimate. For non-Gaussian models, `lgspline` instead uses the usual large-sample Laplace approximation based on the fitted Hessian. For example, for exponential family models, $\mathbf{G}$ is replaced with the inverse negative Hessian of the penalized log-likelihood scaled by the inverse of an exponential dispersion scalar, if present (see 3.2).

2.3 Integrated Curvature Penalty

Regularization for the proposed method uses essentially the equivalent of a quadratic cubic smoothing spline penalty with one predictor (Reinsch, 1967), and a modified quadratic thin-plate spline penalty with multiple predictors (Duchon, 1977; Wahba, 1990), plus a ridge penalty for intercepts and linear terms. Unlike simple ridge regression, this penalty allows higher polynomial terms to be penalized differently than lower-order polynomial terms in a meaningful way that does not rely on heuristics or additional hyperparameters. Theoretically, the penalty is closely related to the norm defining the topology of a Sobolev space $\mathcal{W}_2^2$ (Wahba, 1990). P-spline penalties (Eilers and Marx, 1996) are popular in practice and naturally suited for B-splines due to their banded structure, but by penalizing differences across partitions they cannot be expressed in block-diagonal form, and so are not as amenable to the proposed monomial formulation.

In its simplest form, the regularization scheme proposed for the monomial spline formulation involves constructing a quadratic penalty matrix $\boldsymbol{\Lambda}_k$ for each partition expressible as an elementwise sum of a cubic smoothing spline penalty $\boldsymbol{\Lambda}_s$ weighted by a “wobble penalty” $\lambda_w \geq 0$ and a “flat ridge penalty” for non-spline terms $\boldsymbol{\Lambda}_r$ weighted by $\lambda_r \lambda_w \geq 0$, such that

$\mathbf{\Lambda}_k = \lambda_w(\mathbf{\Lambda}_s + \lambda_r \mathbf{\Lambda}_r)$. Additional predictor-specific and partition-specific penalties can be included as well:

$$\mathbf{\Lambda}_k = \lambda_w \left(\mathbf{\Lambda}_s + \lambda_r \mathbf{\Lambda}_r + \sum_{l=1}^L \lambda_{l,k} \mathbf{\Lambda}_{l,k} \right), \quad (7)$$

for additional penalties $\lambda_{l,k} > 0$ and penalty matrices $\mathbf{\Lambda}_{l,k}$ possibly unique to the k th partition. The full $P \times P$ penalty matrix $\mathbf{\Lambda} = \begin{pmatrix} & & \\ & \ddots & \\ & & \mathbf{\Lambda}_K \end{pmatrix}$ is block-diagonal of $K+1$, $p \times p$ blocks. The parameterization is chosen such that, once derivatives of λ_w are computed, the derivatives for λ_r and $\lambda_{l,k}$ are much faster to compute; details go beyond this paper but are covered in `lgspline` documentation and covered briefly in Supplement C.

The penalty matrix $\mathbf{\Lambda}_s$ serves as the integrated curvature penalty matrix, and can be derived as follows for a single predictor with minimum a and maximum b observed in the dataset:

$$\int_a^b \|f''(t)\|^2 dt = \int_a^b \|\mathbf{x}''(t)^\top \boldsymbol{\beta}_k\|^2 dt = \boldsymbol{\beta}_k^\top \left\{ \int_a^b \mathbf{x}''(t) \mathbf{x}''(t)^\top dt \right\} \boldsymbol{\beta}_k = \boldsymbol{\beta}_k^\top \mathbf{\Lambda}_s \boldsymbol{\beta}_k. \quad (8)$$

For illustration in a single partition with cubic basis $\mathbf{x}(t) = (1, t, t^2, t^3)^\top$ and $\mathbf{x}''(t) = (0, 0, 2, 6t)^\top$:

$$\mathbf{\Lambda}_s = \int_a^b \begin{pmatrix} 0 \\ 0 \\ 2 \\ 6t \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 2 \\ 6t \end{pmatrix}^\top dt = \int_a^b \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 4 & 12t \\ 0 & 0 & 12t & 36t^2 \end{pmatrix} dt = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 4(b-a) & 6(b^2-a^2) \\ 0 & 0 & 6(b^2-a^2) & 12(b^3-a^3) \end{pmatrix}, \quad (9)$$

which is equivalent to the classical cubic smoothing spline penalty of Reinsch (1967) applied to a single partition.

For multiple predictors, an elementwise sum of this operation over all predictors is taken. The second derivative is computed as a total second derivative, accounting for interaction terms, such that for q predictors $\tilde{\boldsymbol{\beta}}_k^\top \mathbf{\Lambda}_s \tilde{\boldsymbol{\beta}}_k = \sum_{j=1}^q \int_{a_j}^{b_j} \left(\frac{\partial^2}{\partial t_j^2} \tilde{f}_k(\mathbf{t}) + \sum_{j \neq j^*} \frac{\partial^2}{\partial t_j \partial t_{j^*}} \tilde{f}_k(\mathbf{t}) \right)^2 dt_j$.

A ridge penalty matrix is included for computational stability of non-spline terms and helps impose positive definiteness of $\mathbf{\Lambda}$, since the second derivative of linear terms and

intercepts is 0, so $\mathbf{\Lambda}_s$ is positive semidefinite. For the single-predictor cubic scenario:

$$\mathbf{\Lambda}_r = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (10)$$

Additional tunable penalties and matrices for partition- or predictor-specific penalization are straightforward to include, for example by isolating the elements of $\mathbf{\Lambda}_s$ that involve only the j th predictor and up-weighting relative to the j^* th predictor. Root mean squared error (RMSE) for constrained leave-one-out cross-validation (Tarpey, 2000) or a constrained variant of generalized cross-validation (Golub, Heath, and Wahba, 1979) are reasonable choices of optimization criteria for tuning the penalties, and are implemented in `lgspline` using gradient-based optimization offered in base R (R Core Team, 2025). See Supplement C for more details.

2.4 Relationship to Other Spline Representations

The practical distinction of the present framework is that the model is fit directly in monomial form. For example, in the cubic single-predictor case the fitted function across all partitions is $\tilde{f}(t) = \sum_{k=0}^K (\tilde{\beta}_{0,k} + \tilde{\beta}_{1,k}t + \tilde{\beta}_{2,k}t^2 + \tilde{\beta}_{3,k}t^3)\mathbb{I}(t \in \mathcal{P}_k)$, sharing the same functional form without reference to knot locations. For B-spline bases the fitted function is $\tilde{f}(t) = \sum_{j=1}^P B_j(t)\tilde{\beta}_j$, where the B_j are B-spline basis functions with compact support, which do not reveal the underlying predictor-response relationship without algebraic rearrangement. For truncated power bases the fitted function may be written as $\tilde{f}(t) = \tilde{\beta}_0 + \tilde{\beta}_1t + \tilde{\beta}_2t^2 + \tilde{\beta}_3t^3 + \sum_{m=1}^K \tilde{\beta}_{m+3}(t - \xi_m)_+^3$, so the representation is explicit but accumulates shifted polynomial terms that directly reference knot locations unless it is rearranged within each partition. Furthermore, the number of coefficients and expansions needed to describe each partition increases, with all estimates needed to describe the predictor-response relationship in at least one partition. Unlike B-splines and the proposed formulation, truncated power bases do not offer compact support.

These three formulations may theoretically span the same spline space, but they differ in numerical behavior and in how directly they expose the local polynomial predictor-response relationship of interest. Each has its own advantage: the monomial parameterization used here fits the local polynomial directly; the B-spline representation favors numerical stability; and the truncated power representation is the simplest to implement without devoted software for imposing constraints or constructing B-spline bases. Section 5.4 examines numerical stability empirically.

3 Computation

3.1 Projection via Transformed OLS

The estimator (5) involves the $P \times P$ matrix $\mathbf{U}$, which is expensive to form when P is large. The central computational device of this article avoids constructing $\mathbf{U}$ entirely by exploiting the block-diagonal structure of $\mathbf{X}$, $\mathbf{G}$, and $\mathbf{A}$. The constrained estimator $\tilde{\boldsymbol{\beta}}$ is obtained as the residuals of an ordinary least-squares problem left-multiplied by $\mathbf{G}^{1/2}$ in four steps:

1. Compute the partition-wise unconstrained estimates $\hat{\boldsymbol{\beta}}$.
2. Transform: $\mathbf{y}^* = \mathbf{G}^{-1/2}\hat{\boldsymbol{\beta}}$, $\mathbf{X}^* = \mathbf{G}^{1/2}\mathbf{A}$.
3. Fit $\mathbb{E}[\mathbf{y}^*] = \mathbf{X}^*\boldsymbol{\gamma}$ for coefficients $\boldsymbol{\gamma}$ by OLS via QR decomposition.
4. Recover $\tilde{\boldsymbol{\beta}} = \mathbf{G}^{1/2}\mathbf{r}^*$ where $\mathbf{r}^* = \mathbf{y}^* - \mathbf{X}^*\hat{\boldsymbol{\gamma}}$.

Back-substitution confirms $\mathbf{G}^{1/2}\mathbf{r}^* = \mathbf{U}\hat{\boldsymbol{\beta}} = \tilde{\boldsymbol{\beta}}$ (Appendix A). Steps 1 and 2, the construction of $\mathbf{A}$, the multiplications by $\mathbf{X}$, and the eigendecomposition of $\mathbf{G}$ used for matrix square roots can all be computed in parallel across $K + 1$ partitions. The most expensive operation is the QR decomposition of the $P \times R$ matrix $\mathbf{X}^*$ in step 3. Since $R \leq P$ typically holds (see Supplement A), this is more efficient than the QR decomposition of the $(N + R) \times P$ matrix used for solving the constrained system as Ezhov, Neitzel, and Petrovic

(2018) describe and compares favorably to the banded operations of B-splines, which are not as straightforwardly parallelizable.

The matrices $\mathbf{G}$, $\mathbf{X}$, and $\mathbf{\Lambda}$ are also memory-efficient; in R, as implemented in `lgspline`, they are stored as a list of $K + 1$ small $p \times p$ matrices rather than collapsed into $P \times P$ block-diagonal form, an advantage that cannot be replicated with truncated power bases.

3.2 Extensions

The same transformed-OLS framework extends naturally beyond the independent Gaussian setting. For GLM responses with working weights $w_i = [V(\mu_i)\{g'(\mu_i)\}^2]^{-1}$, the per-partition scaled inverse information matrix $\mathbf{G}_k$ is now a function of $\boldsymbol{\beta}$ and takes the form $\mathbf{G}_k = (\mathbf{X}_k^\top \mathbf{W}_k \mathbf{X}_k + \mathbf{\Lambda}_k)^{-1}$ with $\mathbf{W}_k = \text{diag}(w_i : i \in \mathcal{P}_k)$. Because $\mathbf{W}$ is diagonal the block-diagonal structure is preserved, and the four-step procedure applies with the weighted $\mathbf{G}_k$. Supplement E can be referenced for more justification.

When the link is non-identity, $\tilde{\mathbf{G}} = \mathbf{G}(\tilde{\boldsymbol{\beta}})$ depends on fitted values through the working weights. At the m th iteration, the approach holds $\hat{\boldsymbol{\beta}}$ fixed at the partition-wise unconstrained solution, recomputes $\mathbf{G}^{(m)}$ using the constrained estimate $\tilde{\boldsymbol{\beta}}^{(m)}$, and re-applies the projection: $\tilde{\boldsymbol{\beta}}^{(m+1)} = \tilde{\mathbf{U}}^{(m)} \hat{\boldsymbol{\beta}}$, iterating until the mean absolute coefficient change falls below a tolerance without worsening in-sample performance. Convergence is typically rapid, usually fewer than 10 iterations. In practice, the computational technique of 3.1 can be used to avoid explicit construction of $\tilde{\mathbf{U}}^{(m)}$ for each iteration. Note that the unconstrained estimates only need to be computed once. The supplement `simple_implementation.R` provides code for demonstrating this with spline logistic regression.

For general inequality constraints $\mathbf{C}^\top \boldsymbol{\beta} \succeq \mathbf{c}$ it is possible that the block-diagonal structure is broken, but this is irrelevant; inequality constraints enter only through the constraint matrix. This is achieved using an active-set method, whereby the inequality constraints that happen to be met using the most recent estimate of $\boldsymbol{\beta}$ are ignored as “inactive”, while inequality constraints that are not yet met get appended to $\mathbf{A}$ as “active” equalities, $\mathbf{A}_{\text{aug}} = [\mathbf{A} \mid \mathbf{C}_A]$, and the same four-step transformed OLS remains unchanged after. The active set is updated by checking primal and dual feasibility until the Karush-Kuhn-Tucker

conditions hold (Nocedal and Wright, 2006).

Handling correlated response requires a more detailed extension of the proposed procedure. When an arbitrary working covariance structure $\mathbf{V}$ is present, $\mathbf{G}^{-1} = \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} + \mathbf{\Lambda}$ is possibly dense, so the block-diagonal factorization used in the independent case is lost naively. But for many covariance structures, the block-diagonal structure can be retained using a Woodbury decomposition. First we separate the scaled information matrix into

$$\mathbf{G}^{-1} = \mathbf{G}_{\text{on}}^{-1} + \mathbf{G}_{\text{off}}^{-1},$$

where $\mathbf{G}_{\text{on}}^{-1}$ collects the within-partition block-diagonal terms and $\mathbf{G}_{\text{off}}^{-1}$ collects the remaining cross-partition terms. When $\mathbf{G}_{\text{off}}^{-1}$ is sparse and low rank with respect to the partition structure, a correction making use of Woodbury's identity preserves the transformed-OLS formulation. This scenario is common when there are many partitions but each correlated cluster overlaps only a few of them. For example, in the special case where each correlated cluster stays within a single partition, $\mathbf{G}_{\text{off}}^{-1} = \mathbf{0}$ and the independent-case block factorization can be recovered exactly.

Writing

$$\mathbf{G}^{-1} = (\mathbf{X}^\top \mathbf{X} + \mathbf{\Lambda}) + \mathbf{X}^\top (\mathbf{V}^{-1} - \mathbf{I}) \mathbf{X},$$

the within-partition portion of the second term is absorbed into corrected per-partition eigendecompositions. The remaining cross-partition term is written using the singular value decomposition as $\mathbf{G}_{\text{off}}^{-1} = \mathbf{E} \mathbf{J} \mathbf{E}^\top$, where $\mathbf{J}$ retains only the s nonzero singular values. Applying the Woodbury identity gives

$$\mathbf{G} = \mathbf{G}_{\text{on}} - \mathbf{G}_{\text{on}} \mathbf{E} (\mathbf{J}^{-1} + \mathbf{E}^\top \mathbf{G}_{\text{on}} \mathbf{E})^{-1} \mathbf{E}^\top \mathbf{G}_{\text{on}} = \mathbf{G}_{\text{on}}^{1/2} \mathbf{F} \mathbf{G}_{\text{on}}^{1/2},$$

with only an $s \times s$ inverse in the correction term. Supplement B shows how to efficiently compute $\mathbf{F}^{1/2}$ and $\mathbf{F}^{-1/2}$ from a QR decomposition of $\mathbf{G}_{\text{on}}^{1/2} \mathbf{E}$, so the transformed-OLS projection is preserved with $\mathbf{G}^{1/2} = \mathbf{G}_{\text{on}}^{1/2} \mathbf{F}^{1/2}$. The main cost remains the per-partition eigendecompositions and the same $P \times R$ QR step as in the independent case, although the construction of the $P \times P$ matrix $\mathbf{F}$ obviates the memory benefit of avoiding explicit construction of $\mathbf{U}$. But with a dense $N \times N$ marginal covariance matrix already $\mathbf{V}$ present, it's worth noting this is likely not a memory bottleneck.

3.3 Bayesian Inference

A Bayesian interpretation of monomial splines is straightforward by interpreting smoothing constraints as an operation applied to the otherwise unconstrained posterior of model parameters. For illustration with the canonical normal-normal conjugate scenario with known σ^2 and the following prior and conditional distributions:

$$\boldsymbol{\beta}|\sigma^2 \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{\Lambda}^{-1}), \quad (11)$$

$$\mathbf{y}|\mathbf{X}, \boldsymbol{\beta}, \sigma^2 \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}). \quad (12)$$

The unconstrained posterior distribution follows the usual normal-normal conjugacy identity:

$$\boldsymbol{\beta}|\mathbf{y}, \mathbf{X}, \sigma^2 \sim \mathcal{N}(\hat{\boldsymbol{\beta}}, \frac{\sigma^2}{N} \mathbf{G}). \quad (13)$$

The posterior distribution under constraint can be expressed as follows for $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$:

$$\boldsymbol{\beta}|\mathbf{y}, \mathbf{X}, \sigma^2, \mathbb{I}_{\mathbf{A}^\top \boldsymbol{\beta} = \mathbf{0}} \equiv \tilde{\boldsymbol{\beta}} + \frac{\sigma}{\sqrt{N}} \mathbf{U} \mathbf{G}^{\frac{1}{2}} \mathbf{z} \equiv \mathbf{U}(\hat{\boldsymbol{\beta}} + \frac{\sigma}{\sqrt{N}} \mathbf{G}^{1/2} \mathbf{z}). \quad (14)$$

This follows from the computationally convenient property $\mathbf{U} \mathbf{G} \mathbf{U}^\top = \mathbf{U} \mathbf{G}$. Thus draws of $\tilde{\boldsymbol{\beta}}$ (which follows a P -dimensional degenerate distribution) can be obtained by simply left-multiplying draws of $\hat{\boldsymbol{\beta}}$ (which follows a non-degenerate P -dimensional distribution) by $\mathbf{U}$. If construction of $\mathbf{U}$ is to be avoided, the same OLS computational technique in 3.1 can be applied by using a draw of $\boldsymbol{\beta}$ from its unconstrained posterior in place of $\hat{\boldsymbol{\beta}}$.

Several choices are made for implementation in `lgspline`, but are not essential to the procedure. A Laplace approximation is implemented for non-Gaussian response, replacing $\mathbf{G}$ with $\tilde{\mathbf{G}}(\tilde{\boldsymbol{\beta}})$ as in 3.2. When inequality constraints are active, elliptical slice sampling (Murray, Adams, and MacKay, 2010) can be used to retain only draws in the feasible region respecting inequality constraints. When correlation parameters are modeled, their uncertainty may be propagated by drawing from a normal approximation centered at the REML estimate, while σ^2 may be drawn from an inverse-gamma distribution derived as the marginal posterior given the observed data and smoothing constraints; in `lgspline` the default prior distribution for σ^2 is *InvGamma*(0, 0). Once σ^2 and correlation parameters are drawn from their posteriors, $\mathbf{G}$ can be re-constructed conditional upon them and in

turn, used to generate posterior draws of coefficients using the approach above. This is demonstrated in Section 5.2.

4 Practical Considerations

The monomial framework is general, but any implementation requires choices regarding partitioning, scaling, how to handle cases in which the smoothness constraints become overdetermined, and handling many non-spline effects simultaneously. We briefly describe the defaults used in `lgspline`; these are not the only valid choices, and the package allows all to be customized.

Knot locations are determined by k -means clustering (MacQueen, 1967) on a standardized predictor matrix, returning centers on the raw scale, with midpoints between neighboring centers serving as partition boundaries. Custom knots, boundaries, and partitioning schemes can be alternatively supplied by the user.

Polynomial basis expansions are scaled by $q_{0.69} - q_{0.31}$, where q_ζ is the ζ th quantile of the expansion, approximately one standard deviation under normality. This robust scaling avoids sensitivity to outliers while keeping higher-order terms on a comparable scale. The penalty matrices and constraints are all scaled by the same values to be consistent. Back-transforming for standard errors and coefficient estimates is computationally negligible, since it only involves rescaling by the same value.

With many partitions or predictors the smoothness constraints can force the model toward a single global polynomial. Dropping second-derivative constraints, first-derivative constraints, or including quartic terms allows a richer fit while overall maintaining perceived smoothness (see Figure 5 for an example).

For models with many non-spline effects and relatively few spline effects, a modified backfitting approach (Hastie and Tibshirani, 1986) alternates between spline-effect estimation using the procedures above and non-spline estimation using standard maximum likelihood. While first-derivative constraints in theory constrain all non-spline effects across partitions to be equivalent (coefficients *are* first derivatives of linear expansions), in practice for q^* non-spline effects there are at least Kq^* redundant coefficients that need to

be estimated is thus computationally inefficient. The modified backfitting does not affect point estimates or standard error estimates but ensures that all non-spline effects remain constant across partitions more efficiently than constrained optimization directly.

5 Results

We report four complementary evaluations. A simulation study across three data-generating processes compares predictive accuracy and runtime against established spline implementations. A pharmacokinetic application emphasizes interpretability and shows how the monomial framework supports shape constraints, structured correlation, and derived parameter inference in a single fit. A parallelism benchmark examines runtime scaling with the number of partitions. A representation-focused analysis reports numerical conditioning of the monomial basis relative to B-spline and truncated power alternatives.

5.1 Simulation Study

We consider three data-generating processes of increasing dimension, each selected to exercise a distinct aspect of multivariate spline fitting. The univariate target is a canonical nonparametric regression benchmark, the bivariate target has sharp local features with genuine two-way interaction, and the four-dimensional target has predictor ranges that differ by several orders of magnitude, stressing numerical conditioning and interaction handling simultaneously.

The first DGP ($q = 1$) is the Doppler function of Donoho and Johnstone (1994) on $[0, 1]$ scaled by 5:

$$f(t) = 5\sqrt{t(1-t)} \sin(2\pi \cdot 1.05 / (t + 0.05)). \quad (15)$$

This function exhibits spatially varying frequency and is a standard benchmark for nonparametric regression methods, since early oscillation is difficult to capture while later oscillations are easier, providing a range of performance that can be realized. The rescaling helps keep choice of noise variance appropriate for benchmarking with other DGPs.

The second DGP ($q = 2$) is the MATLAB `peaks` function (The MathWorks, 2024) on $[-3, 3]^2$:

$$f(t_1, t_2) = 3(1 - t_1)^2 e^{-t_1^2 - (t_2 + 1)^2} - 10\left(\frac{t_1}{5} - t_1^3 - t_2^5\right) e^{-t_1^2 - t_2^2} - \frac{1}{3} e^{-(t_1 + 1)^2 - t_2^2}, \quad (16)$$

a bivariate surface with sharp local features and genuine interaction structure.

The third DGP ($q = 4$) is a four-predictor variant of the Friedman function (Friedman, 1991; Breiman, 1996), the arctangent of an impedance-style ratio:

$$f(t_1, t_2, t_3, t_4) = 10 \cdot \arctan((t_2 t_3 - 1/(t_2 t_4))/t_1), \quad (17)$$

evaluated on the original Friedman ranges $t_1 \in [0, 100]$, $t_2 \in [40\pi, 560\pi]$, $t_3 \in [0, 1]$, $t_4 \in [1, 11]$. The response is scaled by 10 so that its magnitude is comparable to the other two DGPs. The incommensurate predictor ranges produce monomial basis entries spanning many orders of magnitude, which stresses numerical conditioning, and the function itself is non-additive across all four predictors. The original five predictor variant was tested preliminarily but was computationally prohibitive for a full simulation, and $q \leq 4$ spline effects seems most often to be the case in applied research.

Four methods are compared for each DGP. The first is `lgspline`, the proposed monomial spline with a quartic basis and closed-form curvature penalty. The second is a cubic regression spline with cubic smoothing spline penalty (`bs = "cr"`) via `mgcv`. The third method uses thin-plate regression splines (TPRS, tensor-product thin-plate; `bs = "tp"`) via `mgcv`, constructed as tensor products of univariate thin-plate marginals using `s()` for main effects and `te()` for interactions. A thin-plate regression spline constructed on the joint predictor space via `s(t1, ..., tq, bs = "tp")`, rather than as a tensor product of univariate marginals, is more expensive to construct and would likely widen the runtime gaps reported here; neither construction can be rearranged into piecewise polynomial form, but the tensor-product marginal construction was used to keep per-fit times manageable across all scenarios. The fourth method is a truncated quartic polynomial spline with a ridge penalty on the truncated terms, constructed manually. For $q \geq 2$, all models include two-way interactions; for $q = 4$, three-way interactions are also included. For `lgspline`, the parameter K^* controls the number of partitions per predictor $K = K^* q$. For `mgcv`, K^*

is the marginal basis dimension passed to `s()` and `te()`. For the truncated polynomial, K^* is the number of interior knots per predictor. We set $K^* = 16$ for $q = 1$, $K^* = 8$ for $q = 2$, and $K^* = 4$ for $q = 4$, chosen to keep the total model dimension comparable across q while allowing higher resolution in lower dimensions.

Sample sizes are $N \in \{500, 5,000, 50,000\}$ for each q , with Gaussian noise at $\sigma \in \{0.5, 2.0\}$. Two tuning regimes are used: automatic smoothness selection (REML for `mgcv`, exact leave-one-out optimization for `lgspline`, and `mgcv::magic` for the truncated polynomial), and a fixed penalty of 10^{-4} applied to each method’s native penalty structure ($= \lambda_w$ for `lgspline` with $\lambda_r = 0.5$). Each scenario is replicated $M = 24$ times on a 12-core machine, with deterministic per-task seeds. The primary accuracy metric is root mean squared error against the true generating function, $\text{RMSE} = \|\tilde{f} - f\|_2 / \sqrt{N}$. The fixed-penalty regime is not intended as a realistic usage scenario; it is included to isolate per-fit cost from tuning cost and confirm that the runtime comparisons reported below are not artifacts of how each method selects smoothness.

For `lgspline`, an adaptive constraint-relaxation routine is implemented that refits the model without second-derivative constraints, then without first-derivative constraints, in the case the model collapses to a single global polynomial. Relaxation triggered only once across all replications, in a single replication at $q = 4$, $N = 500$, $\sigma = 2.0$ under automatic tuning, where the first-derivative relaxation was invoked. Full constraints were maintained in every other replication, and the additional time for re-fitting is counted in the total fitting time. The truncated polynomial fails at $q = 4$: under the fixed 10^{-4} penalty the design matrix entries from quartic terms of $t_2 \in [40\pi, 560\pi]$ reach magnitudes on the order of 10^{13} , and every one of the 24 replications produced numerical failure (reported as `NaN` in Supplement G). The tuned variant using `mgcv::magic` avoids outright failure but produces RMSE of 3.9-3.95 across every $q = 4$ cell, effectively not fitting the signal at all. These cells are marked with † in Tables 1 and 2 and excluded from the bolding comparison.

Accuracy across DGPs. Table 1 reports mean RMSE under automatic smoothness selection; fixed-penalty results are similar and are deferred to Supplement G. No single method uniformly dominates, and the separation between methods differs across q .

At $q = 1$, `lgspline` achieves the lowest RMSE at $N = 5,000$ and $N = 50,000$ under both noise levels, with the truncated polynomial performing better at $N = 500$ by a narrow margin. For example, at $N = 50,000$ and $\sigma = 0.5$, `lgspline` reaches RMSE 0.666 against 0.688 for the truncated polynomial, 0.692 for Cubic RS, and 0.716 for TPRS. The four methods are within roughly 0.05 RMSE units of each other across most $q = 1$ cells, so the differences are small but consistent, and at this basis dimension none of the methods recovers the first high-frequency oscillation of the Doppler function near $t = 0$ (Figure 1).

At $q = 2$, `lgspline` achieves the lowest RMSE at $N = 500$ under both noise levels (1.102 and 1.163, versus 1.118 and 1.199 for TPRS), but TPRS leads at $N = 5,000$ and $N = 50,000$ for both σ values. The crossover is consistent, since at small N the per-partition flexibility of `lgspline` pays off, but as N grows the smoother global fit of TPRS and Cubic RS begins to emerge as superior. `lgspline` also exhibits larger between-replicate variability than the `mgcv` methods (RMSE SD around 0.07-0.14 versus 0.004-0.05 for Cubic RS and TPRS), reflecting sensitivity of the penalized constrained fit to particular realizations at small partition counts. The truncated polynomial is consistently the worst of the four at $q = 2$, with RMSE around 1.30-1.35. Overall, the methods at $q = 2$ are broadly comparable in accuracy, with the choice between them depending more on N than on any systematic advantage of one basis.

At $q = 4$, Cubic RS achieves the lowest RMSE at $N = 500$ under both noise levels (0.738 and 1.009 versus 0.780 and 1.201 for `lgspline`), and `lgspline` appears superior at $N = 5,000$ and $N = 50,000$ by substantial margins. At $N = 50,000$ and $\sigma = 0.5$, `lgspline` reaches RMSE 0.543 against 0.790 for Cubic RS and 0.909 for TPRS; the truncated polynomial fails altogether at RMSE near 3.95. The small- N loss to Cubic RS at $q = 4$ reflects the same between-replicate instability that `lgspline` shows at $q = 2$, compounded by the small ratio of observations to partitions: $N = 500$ across $K = 16$ partitions gives only about 30 observations per partition. As N grows this ratio improves and `lgspline` clearly dominates.

This interpretation should be made with caution. The four methods differ in more than just basis, and the comparison is not a like-for-like benchmark in the strict sense.

Smoothness is selected by exact leave-one-out for `lgspline`, REML for the `mgcv` methods, and `mgcv::magic` generalized cross-validation for the truncated polynomial; `lgspline` uses quartic expansions as its default and the truncated polynomial is given quartic expansions to match, while the `mgcv` constructions use their native cubic bases; and K^* means partitions per predictor for `lgspline`, marginal basis dimension for `mgcv`, and interior knots per predictor for the truncated polynomial, so the three quantities align only loosely. Exactly matching model formulations across these frameworks is not straightforward, and some of the variability in performance is likely attributable to these differences rather than to any fundamental property of the bases. The broader conclusion does not depend on the alignment being exact, however; there is little evidence here suggesting a consistent accuracy or speed disadvantage from working in monomial form, performance is comparable (if by seconds slightly slower) at small N and small q , and the efficiency advantages appear to grow with N and possibly q .

Timing. Figure 2 presents wall-clock time as a function of sample size on a log scale, faceted by q and noise level, with methods distinguished by color and tuning regimes by line type. Three patterns can be noticed. First, at small N and low q , Cubic RS is the fastest method by a comfortable margin; `lgspline` pays a fixed overhead for its eigendecomposition and QR steps that dominates when N is small, although the total cost to the user is about a second. Second, as N grows the ordering shifts: at $q = 1$, $N = 50,000$ under tuning at $\sigma = 0.5$, `lgspline` completes in 5.7 seconds, edging ahead of Cubic RS at 6.5 seconds, with TPRS about three times slower at 18.4 seconds; overall, still similar. At $q = 2$, $N = 50,000$, Cubic RS retains a modest edge (50.2 seconds versus 62.0 for `lgspline` and 53.8 for TPRS). Third, at $q = 4$ the gap becomes dramatic and consistent across both noise levels.

Under automatic tuning at $q = 4$, $N = 50,000$, $\sigma = 0.5$, `lgspline` completes every replication in 171 seconds on average, while Cubic RS averages 1,850 seconds and TPRS averages 937 seconds, yielding speedups of roughly $11\times$ and $5.5\times$. At $\sigma = 2.0$ the corresponding tuned timings are 1,750 seconds (Cubic RS) and 1,000 seconds (TPRS) against `lgspline`'s 176 seconds, giving speedups of $10\times$ and $5.7\times$. Full numerical timings appear

in Table 2. The fixed-penalty regime in Figure 2 confirms that these advantages persist when the tuning loop is removed, so the speedups are not an artifact of simply how each method selects smoothness parameters, but shows that the computational efficiency of the monomial formulation is superior with larger N and q .

Taken together, the simulation shows that **lgspline** tends to be the most accurate method at $q = 1$ (outside the smallest sample size) and at moderate-to-large N for $q = 4$, broadly comparable to the **mgcv** methods at $q = 2$, and prone to some instability at small N relative to partition count. The computational advantages become substantial once q grows: at $q = 4$, **lgspline** is the only method that finishes every replication at $N = 50,000$ in under three minutes on average, and it is simultaneously the most accurate. The truncated polynomial is competitive at $q = 1$ but fails at $q = 2$ in accuracy and collapses entirely at $q = 4$, supporting the view that the block-diagonal constrained reformulation is responsible for the stability of the **lgspline** approach on multivariate domains.

Table 1: RMSE against f_{true} (mean (SD)) under automatic smoothness selection over 24 replications. Lowest RMSE in each column is **bold**. Cells marked $\dagger$ at $q = 4$ indicate truncated-polynomial failure (infeasible RMSE from conditioning breakdown) and are excluded from bolding. Fixed-penalty results appear in Table H.1.

σ	Method	$q = 1$			$q = 2$			$q = 4$		
		$N=500$	$N=5k$	$N=50k$	$N=500$	$N=5k$	$N=50k$	$N=500$	$N=5k$	$N=50k$
0.5	lgspline	0.680 (0.058)	0.686 (0.047)	0.666 (0.049)	1.102 (0.137)	1.145 (0.110)	1.246 (0.072)	0.780 (0.106)	0.546 (0.046)	0.543 (0.039)
	Cubic RS	0.714 (0.068)	0.703 (0.027)	0.692 (0.013)	1.145 (0.050)	1.138 (0.015)	1.139 (0.004)	0.738 (0.077)	0.781 (0.028)	0.790 (0.009)
	TPRS	0.714 (0.045)	0.724 (0.010)	0.716 (0.003)	1.118 (0.047)	1.128 (0.013)	1.127 (0.004)	0.816 (0.083)	0.899 (0.032)	0.909 (0.009)
	Trunc. poly	0.673 (0.064)	0.696 (0.021)	0.688 (0.012)	1.300 (0.062)	1.316 (0.015)	1.317 (0.005)	3.899 † (0.142)	3.944 † (0.062)	3.946 † (0.015)
2.0	lgspline	0.782 (0.068)	0.682 (0.058)	0.668 (0.043)	1.163 (0.090)	1.201 (0.078)	1.264 (0.065)	1.201 (0.098)	0.646 (0.038)	0.556 (0.032)
	Cubic RS	0.809 (0.058)	0.710 (0.039)	0.698 (0.014)	1.222 (0.043)	1.163 (0.012)	1.139 (0.004)	1.009 (0.086)	0.833 (0.032)	0.794 (0.010)
	TPRS	0.803 (0.039)	0.730 (0.012)	0.717 (0.003)	1.199 (0.041)	1.142 (0.011)	1.126 (0.003)	1.020 (0.086)	0.935 (0.038)	0.908 (0.010)
	Trunc. poly	0.774 (0.055)	0.700 (0.037)	0.694 (0.013)	1.344 (0.048)	1.327 (0.013)	1.317 (0.005)	3.914 † (0.144)	3.940 † (0.042)	3.942 † (0.015)

5.2 Application to Pharmacokinetics

We illustrate several features of **lgspline** simultaneously on the Theophylline pharmacokinetic dataset available in base R (Boeckmann, Sheiner, and Beal, 1994; R Core Team, 2025), comprising 132 observations from 12 subjects each measured at 11 irregularly spaced

Table 2: Mean wall-clock time in seconds under automatic smoothness selection over 24 replications (SD in parentheses). Fastest method in each column is **bold**. Cells marked † at $q = 4$ indicate truncated-polynomial failure (infeasible RMSE) and are excluded from the bolding comparison. Fixed-penalty timings appear in Table H.2.

σ	Method	$q = 1$			$q = 2$			$q = 4$		
		500	5k	50k	500	5k	50k	500	5k	50k
0.5	lgspline	0.7 (0.1)	1.5 (0.4)	5.7 (1.4)	3.2 (0.9)	7.1 (3.0)	62.0 (29.8)	33.9 (16.1)	31.3 (6.6)	171 (79)
	Cubic RS	0.2 (0.1)	0.5 (0.2)	6.5 (1.8)	0.7 (0.2)	4.2 (1.2)	50.2 (15.0)	39.2 (20.7)	281 (145)	1850 (405)
	TPRS	0.3 (0.2)	2.1 (0.4)	18.4 (4.4)	1.1 (0.5)	5.0 (0.7)	53.8 (8.8)	30.8 (8.6)	213 (95)	937 (218)
	Trunc. poly	0.0 (0.1)	0.1 (0.2)	0.1 (0.0)	0.1 (0.1)	0.0 (0.0)	0.2 (0.0)	0.1 [†] (0.0)	0.0 [†] (0.0)	0.2 [†] (0.1)
2.0	lgspline	0.8 (0.1)	1.5 (0.4)	5.5 (1.4)	3.3 (1.2)	6.2 (2.2)	59.5 (25.6)	34.7 (12.6)	34.6 (10.8)	176 (91)
	Cubic RS	0.2 (0.2)	0.6 (0.2)	6.5 (1.9)	0.5 (0.2)	2.4 (0.3)	55.9 (13.3)	22.1 (9.6)	222 (113)	1750 (494)
	TPRS	0.3 (0.2)	1.9 (0.3)	16.9 (4.4)	1.3 (0.6)	5.9 (1.1)	60.8 (10.2)	26.8 (9.6)	256 (110)	1000 (349)
	Trunc. poly	0.0 (0.1)	0.1 (0.1)	0.2 (0.1)	0.1 (0.2)	0.0 (0.1)	0.2 (0.0)	0.1 [†] (0.0)	0.0 [†] (0.0)	0.2 [†] (0.1)

time points over approximately 24 hours post-dose. The aim is not to optimize predictive performance but to demonstrate that the monomial framework supports biologically motivated model construction, structured covariance, simultaneous shape constraints, and direct extraction of derived pharmacokinetic parameters following standard non-compartmental analysis (Gabrielsson and Weiner, 2016) with fully propagated uncertainty in a single integrated fit.

Serum concentration was modeled as a quadratic spline in time with a linear effect of weight-normalized dose and a normalized dose-by-time quadratic interaction, using $K = 10$ interior knots placed between consecutive observed time points, yielding 11 partitions. Model structure was chosen to reflect biological plausibility and to illustrate features of **lgspline** rather than selected by AIC or cross-validation. Specifically, within each partition the model takes the form

$$\tilde{f}(\text{Time}, \text{Dose}) = \beta_0 + \beta_1 \text{Time} + \beta_2 \text{Time}^2 + \beta_3 \cdot \text{Dose} + \beta_4 \cdot \text{Dose} \cdot \text{Time}^2,$$

with Time in hours since last dose, and Dose as weight-normalized dosage of Theophylline in $mg/L/kg$, and the Dose-Time interaction is included as a quadratic-linear interaction term. The interaction allows dose to modulate the curvature of the concentration-time profile in a manner that is both interpretable and non-constant across the time domain

through a single parameter. By excluding a linear interaction $\text{Time} * \text{Dose}$, the initial rate of absorption does not depend on the initial Dose while the initial acceleration does, which is in alignment with general understanding of pharmacokinetics. Only continuity and first-derivative smoothness were imposed; second-derivative constraints were excluded, which is typical for a quadratic spline.

Within-subject correlation was modeled via a customized tridiagonal exchangeable correlation matrix with a common estimated parameter ρ , such that only neighboring partitions with respect to time are correlated (the first timepoint is correlated with the second timepoint for each subject, but not the third timepoint, nor between subjects). This structure was chosen to ensure sparsity of the off-block-diagonal elements of the marginal variance-covariance matrix while capturing the dominant serial dependence between temporally adjacent measurements. The correlation parameter was estimated by maximizing the REML profile likelihood using the structured covariance extension described in Section 3.2. Three simultaneous inequality constraints were imposed for biological plausibility: a non-positive first derivative with respect to time for $\text{Time} > 5$ hours, enforcing monotone elimination; a non-negative second derivative for $\text{Time} > 5$ hours, enforcing concavity during terminal elimination; and non-negative fitted values for $\text{Time} > 5$. The time threshold of 5 hours was chosen to exclude the absorption and early distribution phase.

A notable feature of the fitted model is that the quadratic interaction of dose and time is held constant across partitions. This is a natural consequence of the smoothness constraints and this specific dataset rather than a modeling artifact or computational issue, and is convenient in that it reduces the number of partitions with differing functional relationships. This phenomenon can be avoided by including higher-order polynomial terms and the linear interaction term, but is argued here to be an interpretable advantage of the simple formulation, and likely attractive to applied researchers.

The leftmost partition, covering the absorption phase, yields the closed-form quadratic polynomial

$$\tilde{f}(\text{Time}, \text{Dose}) = -1.3388 + 7.0434 \text{ Time} + 7.7965 \text{ Time}^2 + 0.2941 \cdot \text{Dose} + 0.001 \cdot \text{Dose} \cdot \text{Time}^2.$$

The first derivative at $\text{Time} = 0$ is $\tilde{f}'(0, \text{Dose}) = 7.0434 \text{ mg/L/h}$, giving the estimated

initial rate of absorption directly from the monomial coefficients, which does not depend on dosage. The second derivative at Time = 0 is $\tilde{f}''(0, \text{Dose}) = 2 \times 7.7965 + 2 \times 0.001 \times \text{Dose} = 15.593 + 0.002 \cdot \text{Dose} \text{ mg/L/h}^2$, representing the initial acceleration of absorption, which is modeled to increase linearly with dosage. Both quantities are available as closed-form expressions in every partition and can be evaluated at any covariate value without post-hoc algebra by someone with elementary calculus, which would not be immediately possible with a truncated power or B-spline basis.

For posterior inference, let $\tilde{f}^{(m)}$ denote the fit concentration-time curve at dose d implied by the posterior draw m th of the coefficients $\tilde{\boldsymbol{\beta}}^{(m)}$. Each of the $M = 1,000$ posterior draws is generated following Section 3.3, satisfying smoothness constraints without imposing inequality constraints (e.g. elliptic slice sampling was not used). Draws of the correlation parameter ρ were obtained from a normal approximation on the $\tanh^{-1}$ scale, using the REML point estimate as the mean and the inverse approximate second derivative from the BFGS algorithm used to optimize REML as the covariance, then back-transformed. The variance parameter σ^2 was drawn from its marginal posterior density of $\sigma^2 | \mathbf{y}, \mathbf{X} \sim \text{InvGamma}(\frac{1}{2}(N - \text{tr}(\mathbf{H})), \frac{1}{2}(N - \text{tr}(\mathbf{H}))\tilde{\sigma}^2)$, with $\tilde{\sigma}^2$ as defined in 3.1 which follows from `lgspline` defaults. The six pharmacokinetic summaries extracted from each posterior draw are listed in Table 3; each follows standard non-compartmental definitions (Gabrielsson and Weiner, 2016) and are widely used in practice. As each partition is a polynomial, all derivatives and function evaluations needed for these summaries are available directly from $\tilde{f}^{(m)}(t; d)$ without basis conversion. Area under the curve (AUC) at 24 hours was computed using Gauss–Legendre quadrature applied to the posterior concentration curve. Draws producing a negative predicted concentration at Time = 24 hours, corresponding to draws for which $\text{AUC}_{0-24} \leq 0$ or for which the $\text{AUC}_{0-24}/\text{AUC}_{\infty}$ ratio is at least 1, were excluded as biologically implausible. Fewer than 1.5% of draws fell into this category.

Figure 3 summarizes the analysis visually. The left panel shows the fitted concentration-time curve at mean normalized dose with a pointwise 95% posterior band, with observed data overlaid. The curve rises sharply through the absorption phase and transitions

Table 3: Posterior-draw definitions of the pharmacokinetic summaries computed from the fitted concentration-time curve $\tilde{f}^{(m)}(t; d)$ at dose d .

Name	Symbol	Description	Formula
Time to peak concentration	$t_{\max}^{(m)}$	Time at which the fitted concentration attains its maximum on $[0, 24]$	$\arg \max_{0 \leq t \leq 24} \tilde{f}^{(m)}(t; d)$
Peak concentration	$C_{\max}^{(m)}$	Maximum fitted concentration	$\tilde{f}^{(m)}(t_{\max}^{(m)}; d)$
Half-maximum time	$t_{1/2}^{(m)}$	First post-peak time at which the fitted concentration falls to one half of $C_{\max}^{(m)}$	$\inf\{t > t_{\max}^{(m)} : \tilde{f}^{(m)}(t; d) = C_{\max}^{(m)}/2\}$
Terminal elimination rate constant	$\lambda_z^{(m)}$	Log-linear elimination slope implied by the peak and half-maximum times	$\frac{\log(0.5)}{t_{\max}^{(m)} - t_{1/2}^{(m)}}$
Exposure through 24 hours	$\text{AUC}_{0-24}^{(m)}$	Area under the fitted concentration-time curve on the observed window	$\int_0^{24} \tilde{f}^{(m)}(t; d) dt$
Exposure fraction observed by 24 hours	$\text{AUC}_{\text{ratio}}^{(m)}$	Fraction of extrapolated total exposure captured by 24 hours	$\frac{\text{AUC}_{0-24}^{(m)}}{\text{AUC}_{0-24}^{(m)} + \tilde{f}^{(m)}(24; d)/\lambda_z^{(m)}}$

smoothly to monotone concave decay after Time = 5 hours, with the shape constraints eliminating non-monotone artifacts that would otherwise appear during the terminal phase. The partition-specific polynomial equations in the caption key provide a standalone, fully interpretable description of the fitted model. The right panel shows the posterior densities of the six pharmacokinetic summaries extracted from the $M = 1,000$ draws, conveying the shape of the posterior rather than only its quantiles. Heavy tails on AUC_{0-24} , $t_{1/2}$, and λ_z reflect the sensitivity of terminal-phase extrapolation to the estimated correlation parameter and the short observation window relative to full elimination.

The point estimate of the correlation parameter is 0.519, and a 95% confidence interval based on the inverse Hessian of the BFGS optimization implemented in the `optim` function of base R (R Core Team, 2025) for REML gives bounds of 0.502 to 0.535.

Table 4 reports posterior quantiles and 95% credible intervals at normalized Dose = 5 mg/kg. The expected time to peak concentration has a posterior median of 1.37 hours with a 95% CI of 1.13 to 3.07, consistent with rapid absorption. The expected maximum concentration has a median of 8.70 mg/L with a 95% CI of 7.47 to 10.15. Elimination

Table 4: Posterior pharmacokinetic summary distributions from `lgspline` ($M = 1000$ draws total, 985 draws retained after excluding biologically implausible draws, normalized Dose = 5 mg/kg).

Parameter	Min	2.5%	25%	Median	75%	97.5%	Max
$t_{\max}$ (hours)	0.74	1.13	1.31	1.37	1.49	3.07	5.35
$C_{\max}$ (mg/L)	6.92	7.47	8.21	8.70	9.18	10.15	11.00
$t_{1/2}$ (hours)	6.03	8.28	9.92	10.76	11.63	14.37	24.65
λ_z (h^{-1})	0.029	0.054	0.069	0.075	0.083	0.109	0.147
AUC_{0-24} (mg·h/L)	1.18	16.41	70.55	95.44	123.18	177.07	228.73
$\text{AUC}_{0-24}/\text{AUC}_{\infty}$	0.04	0.45	0.74	0.80	0.85	0.94	0.99

half-life has a median of 10.76 hours with a 95% CI of 8.28 to 14.37, and the terminal elimination rate constant has a median of 0.075 h^{-1} with a 95% CI of 0.054 to 0.109, indicating a terminal decline that is appreciable but not abrupt over the 24-hour follow-up. Total drug exposure has a median AUC_{0-24} of 95.4 mg·h/L with a 95% CI of 16.4 to 177.1. The wide interval reflects the within-subject correlation contribution to the effective variance of the constrained estimate, and uncertainty involved with a small sample size and complicated non-linear predictor-response relationship. The $\text{AUC}_{0-24}/\text{AUC}_{\infty}$ ratio has a median of 0.80 with a 95% CI of 0.45 to 0.94, suggesting that for the majority of subjects a meaningful fraction of total exposure occurs beyond 24 hours.

The combination of user-specified structured within-subject correlation, simultaneous arbitrary shape constraints, and a biologically motivated model built directly in monomial form is not, to our knowledge, jointly available in a single fit in existing R spline software. These results demonstrate that the monomial framework recovers clinically interpretable PK summaries with fully propagated posterior uncertainty in a single integrated fit, requiring no post-hoc algebraic basis conversion nor assumption of an underlying compartmental model.

To assess the contribution of the normalized dose-by-time quadratic interaction term, a nested model was fit under the additional linear constraint that the interaction coefficient is

exactly zero in every partition, holding the penalty and covariance structure fixed at their full-model estimates. This is accomplished by simply re-fitting the model but with tuning parameters and correlation fixed, and with an additional 11 columns appended to $\mathbf{A}$ coding the additional constraints. The 11 constraints reduce to only one degree of freedom after QR pivot, the difference in rank between $\mathbf{A}$ of the full model and nested model, because the smoothness constraints already force equivalence across partitions. An analogous test with B-spline or truncated power bases would require constraining many coefficients distributed across partitions while preserving the penalty structure, which is not straightforward in most software.

A Wilks-type likelihood ratio statistic of 0.61 on 1 degree of freedom yields a nominal p -value of 0.44 under a χ^2_1 reference. This should be interpreted cautiously as classical asymptotics may not be fully satisfied under active inequality constraints, and the penalty and correlation parameters are fixed rather than re-optimized under the null, so the reference distribution is only approximate. Taken at face value the test provides little evidence against the reduced model, suggesting that the normalized dose-by-time quadratic interaction contributes little beyond the main effects given this sample size and correlation structure. This makes sense because each participant was given the same absolute dose and only the weight differed between the participants, which could be captured in part by intrasubject correlation. Deriving the exact or asymptotic null distribution under these conditions, or calibrating it by parametric bootstrap, is left for future research. The practical point is that the monomial framework permits nested likelihood comparisons under an identical penalty and covariance structure through the same constraint machinery used for enforcing smoothness and shape constraints, without requiring basis reparameterization.

5.3 Evaluating Parallelism

To assess the computational behavior of the block-diagonal structure under parallelism, `lgspline` was fit to the Maunga Whau volcano dataset (R Core Team, 2025), an 87×61 grid yielding $N = 5,307$ observations with $q = 2$ predictors (easting and northing) and elevation as the response. Five partition counts were considered: $K + 1 \in \{50, 100, 250, 500, 1000\}$.

Each configuration was fit five times with and without a 12-core parallel cluster, using a fixed penalty, no derivative constraints, and quadratic interactions. Models were not tuned, because different tuning runs have different numbers of iterations and introduce variance not attributable to parallelism. Parallel processing was applied to neighbor determination for each partition, smoothing constraint setup, and the eigenvalue decompositions used to compute the square root and inverse square root of $\mathbf{G}$.

Figure 4 shows the measured speedup (sequential time divided by parallel time) as a function of partition count. At $K + 1 = 50$, parallel overhead dominates and sequential fitting is faster by a factor of roughly 5 (speedup 0.18 $\times$). The crossover occurs between $K + 1 = 100$ and $K + 1 = 250$, with parallel fitting first becoming beneficial at the latter (1.25 $\times$). The speedup climbs to 1.50 $\times$ at $K + 1 = 500$ and plateaus at 1.48 $\times$ at $K + 1 = 1000$, which might reflect the QR decomposition of $\mathbf{A}$ and $\mathbf{G}^{1/2}\mathbf{A}$ as rate-limiting steps. RMSE (Table 5) decreases monotonically from 1.49 at $K + 1 = 50$ to 0.85 at $K + 1 = 1000$, with meaningful reduction already achieved by $K + 1 = 250$, confirming that the model did not collapse or perform worse on held-out data with more partitions. The parallel gains are modest here, but might be better realized with an implementation using graphics processing units (GPUs) or other highly parallelizable hardware.

Figure 5 shows an illustrative tuned fit with $K + 1 = 250$ for context.

Table 5: Wall-clock time in seconds for `lgspline` on the volcano dataset (87×61 grid, $N = 5,307$, $q = 2$), fixed penalty, mean $\pm$ SD over 5 replications.

$K+1$	Sequential		Parallel (12 cores)		Speedup		RMSE	
50	0.2	(0.0)	2.8	(3.6)	0.18	(0.13)	1.49	(0.08)
100	1.2	(0.0)	3.4	(2.7)	0.53	(0.27)	1.16	(0.06)
250	17.3	(0.2)	14.6	(4.3)	1.25	(0.27)	0.96	(0.05)
500	158.5	(0.4)	105.7	(6.3)	1.50	(0.08)	0.89	(0.04)
1000	1491.0	(16.6)	1009.1	(7.7)	1.48	(0.02)	0.85	(0.02)

5.4 Basis Representation and Stability

The preceding results demonstrate that the monomial basis yields competitive predictive accuracy across $q = 1, 2, 4$ despite the well-known conditioning concerns associated with monomial expansions. We close with a focused diagnostic that quantifies the conditioning tradeoff directly and shows that it is manageable at the scales tested.

We evaluate the numerical stability and predictive performance of `lgspline` across wiggle penalty levels $\lambda_w \in \{0, 10^{-10}, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}, 1\}$ with a fixed ridge regularization $\lambda_r = 1$ applied to the null space of the curvature penalty. Data are generated from a smooth bivariate function on $[0, 1]^2$ with $N = 1000$ observations and additive Gaussian noise $\sigma = 1$:

$$\begin{aligned} \tilde{t}_1 &= t_1 - 1/2, \quad \tilde{t}_2 = t_2 - 1/2 \\ f(t_1, t_2) &= 2 + 0.8 \sin(4\tilde{t}_1) + 0.7 \cos(4\tilde{t}_2) + 0.6 \tilde{t}_1 \tilde{t}_2 \\ &\quad - 0.4 \tilde{t}_1^2 + 0.3 \tilde{t}_2^2 + 0.15 \tilde{t}_1^3 - 0.10 \tilde{t}_2^3 \end{aligned} \tag{18}$$

The support is partitioned using $K + 1 = 4$ partitions. Results are based on $M = 250$ Monte Carlo replicates. We report mean and standard deviation of partition-wise condition numbers of the correlation matrices $\kappa(\mathbf{G}_k)$ for the monomial formulation, along with the condition number of the analogous correlation matrix obtained from B-spline and truncated power setups. RMSE from the true data-generating function is also reported. B-spline and truncated power models are fit on the same design but are not penalized, so only their unpenalized results appear.

The monomial basis is substantially more stable than the global truncated power basis (by roughly four orders of magnitude even at $\lambda_w = 0$), and roughly four orders of magnitude less stable than B-splines at $\lambda_w = 0$. Moderate penalization ($\lambda_w \approx 10^{-2}$) reduces the monomial condition number by nearly three orders of magnitude, closing the gap with B-splines to about two orders of magnitude while yielding a slight improvement in RMSE. The simulation results of Section 5.1 confirm that the conditioning disadvantage does not translate into accuracy loss at scales and penalty structures considered, even at $q = 4$ where the truncated polynomial often collapses due to conditioning.

Table 6: Mean partition condition number and RMSE for `lgspline` across wiggle penalty levels over $B = 250$ replicates. Values are mean (SD). B-spline and truncated power entries are reported only at $\lambda_w = 0$.

λ_w	Mean $\kappa(\mathbf{G}_k)$	RMSE
0	1.55×10^7 (3.07×10^6)	0.1345 (0.02410)
10^{-10}	1.51×10^7 (3.09×10^6)	0.1350 (0.02436)
10^{-8}	1.54×10^7 (3.30×10^6)	0.1321 (0.02496)
10^{-6}	1.51×10^7 (3.21×10^6)	0.1354 (0.02420)
10^{-4}	6.93×10^6 (9.88×10^5)	0.1316 (0.02447)
10^{-2}	4.03×10^5 (2.37×10^4)	0.1285 (0.01898)
1	4.65×10^4 (6.05×10^3)	0.2145 (0.01316)
B-spline	1.79×10^3 (3.12×10^2)	0.2115 (0.03006)
Trunc. power	1.36×10^{11} (1.19×10^{10})	0.1710 (0.02657)

6 Discussion

The main contribution of this article is a computational strategy for fitting penalized polynomial splines in monomial form using constrained optimization. The transformed OLS formulation exploits the block-diagonal partition structure for both memory savings and faster computation, while also creating a natural path to parallel execution. This computational convenience is retained under GLM weights, inequality constraints including shape constraints, and certain correlated settings.

Relative to truncated polynomial splines, the proposed method keeps the representation local. Each partition depends on only p coefficients rather than potentially referencing the full P -dimensional coefficient vector, and the full dense linear system never needs to be assembled. Relative to B-spline methods, the method trades some numerical conditioning for direct interpretability. The unpenalized monomial basis is roughly four orders of magnitude less well conditioned than B-splines, which is a real cost; moderate penalization closes this gap to about two orders of magnitude (Section 5.4), and the $q = 4$ simulation confirms that this residual gap does not translate into accuracy loss at the tested scales, even when the unpenalized truncated polynomial collapses entirely due to conditioning. The monomial form allows equations from fitted models to be read, checked, and differentiated directly

without basis conversion, and as shown in the pharmacokinetic application of Section 5.2, enables certain combinations of modeling features that are awkward or difficult to assemble in other spline frameworks.

The simulation study of Section 5.1 shows that **lgspline** is the most accurate method at $q = 1$ outside the smallest sample size, the most accurate method at $q = 4$ once $N \geq 5,000$, and broadly comparable to the **mgcv** methods at $q = 2$. The method does show small- N instability at high partition counts: at $q = 4$, $N = 500$ Cubic RS is more accurate under both noise levels, and **lgspline** exhibits consistently larger between-replicate standard deviations than the **mgcv** methods at $q = 2$. As the ratio of observations to partitions grows this instability weakens and the accuracy ordering cleanly favors **lgspline**. On runtime, Cubic RS is the fastest method for small N and low q because of **lgspline**'s fixed overhead from eigendecompositions and the constraint QR, but the difference is less than a second on average for $q = 1$ and about 12 seconds at worst when $q = 2$ and $N = 50,000$. By $N = 50,000$ at $q = 1$ **lgspline** is faster, and at $q = 4$ with $N \geq 5,000$ the block-diagonal structure produces speedups of roughly $5\times$ - $11\times$ over the **mgcv** methods. The fixed-penalty regime confirms these speedups are not simply artifacts of how each method handles smoothness selection, but inherent to how they are fit. The truncated polynomial is competitive at $q = 1$ but fails at $q = 2$ in accuracy and collapses entirely at $q = 4$, supporting the view that the block-diagonal constrained reformulation rather than the polynomial degree is responsible for the stability of the **lgspline** approach on multivariate domains. The pharmacokinetic application shows the practical value of the method for carefully constructed parametric models of predictor-response relationships in an interpretable and immediately useful form. The volcano example highlights the parallelism ceiling imposed by the constraint QR operation, hinting that differing hardware is needed to fully realize the parallel processing advantages of the method.

There are clear limitations to the proposed method that should be addressed. First, smoothness and shape constraints are satisfied only to the numerical tolerance of the computing environment. Constraint enforcement can be checked informally by computing the maximum absolute value of $\mathbf{A}^\top \tilde{\boldsymbol{\beta}}$, which in theory should be zero and in practice should

be very small (or $\mathbf{A}^\top \tilde{\boldsymbol{\beta}} - \mathbf{c}$ for non-zero constraints). Second, overconstrained models can force the spline to reduce to a single polynomial shared across all partitions, which becomes more common with many spline predictors; the adaptive relaxation implemented in `lgspline` triggered only once across the full simulation study, but was designed to handle this case when it arises. Third, inequality constraints imposed via the active-set procedure are enforced at observed data points only, unlike the global pointwise enforcement of B-spline shape-constrained implementations such as `scam` (Pya and Wood, 2015). Whether inequality constraint enforcement extends to unobserved regions of the predictor support depends on the data layout and polynomial degree, and additional user-specified constraint points may be needed to guarantee coverage in sparse regions. Fourth, in high dimensions the number of partition neighbors grows rapidly, inflating the constraint system and degrading some of the computational advantages of the block-diagonal structure; an additive decomposition in which smoothness constraints operate independently per predictor would reduce this burden but is not currently implemented. Fifth, penalization is effectively necessary when the partition count is large relative to the number of observations; replicating a smoothing spline with one partition per observation, for instance, introduces p free coefficients per observation, and the unconstrained per-partition fits required by the transformed-OLS procedure become numerically degenerate without sufficient regularization. This is also what drives the between-replicate variability of `lgspline` observed at $q = 2$ and the small- N accuracy loss at $q = 4$, $N = 500$, such that as the observations-per-partition ratio increases, stability decreases. Sixth, when numerical instability does occur, the results may appear as a failure of constraint enforcement (for example, a disconnected polynomial curve or inequality constraints that appear to be ignored). Cases in which constraints fail to be enforced typically signal that a model is too complex or overparameterized for the data available, and so the failure is itself informative. Finally, polynomials may not be easily interpretable compared to linear terms, and interpreting a piecewise fit requires more care than a single global parametric form, but this is a fundamental limitation of polynomial splines and is not unique to the monomial formulation. Interpretability is arguably improved over truncated bases, which tend to require many

more coefficients to describe predictor-response relationships using formulas that explicitly contain knot locations, and B-splines using basis functions that are difficult to interpret without further algebraic manipulation.

We argue in favor of using the proposed monomial spline in practice, but it has intrinsic value as an educational device alone. Not only is it a particularly rich application of constrained optimization to statistics, but many students may appreciate seeing a polynomial in a familiar form before moving on to more complicated basis constructions, which is in agreement with Ezhov, Neitzel, and Petrovic (2018). Even a reader who prefers B-splines for their numerical properties may find the monomial formulation useful as an educational complement, particularly through the single-predictor base R reference implementation provided in Supplement H.

Potential future work includes quantile regression and location-scale extensions in the spirit of `qgam` (Fasiolo, Wood, Zaffran, Nedellec, and Goude, 2021) and `gamlss` (Stasinopoulos and Rigby, 2007), more aggressive parallel and GPU implementations targeting parallelism, and the derivation of null distributions for likelihood-ratio tests under active inequality constraints and fixed penalty parameters. More broadly, the same transformed constrained-OLS framework may be useful beyond spline fitting whenever a problem has block-diagonal design structure and linear constraints. Software implementing all procedures described here is available in `lgspline` for R on CRAN.

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Supplementary Materials

Code: R scripts reproducing all simulations and applications, available at <https://github.com/matthewlouisdavisBioStat/lgspline>. Reported results correspond to `lgspline` version 1.0.2 on CRAN. The scripts are named `performance_sim.R` (simulation study of Section 5.1), `nca_analysis.R` (pharmacokinetic application of Section 5.2), `volcano_timing.R` (parallelism benchmark of Section 5.3), `stability_sim.R` (basis conditioning analysis of Section 5.4), and `simple_implementation.R` (the single-predictor base R reference implementation of Supplement I).

Supplement A: Multivariate basis example and constraint counting.

Supplement B: Woodbury factorization for structured covariance (full derivation of Section 3.2).

Supplement C: Penalty construction and tuning.

Supplement D: REML for correlation parameters.

Supplement E: GLM consistency and asymptotic normality of the constrained estimator.

Supplement F: Approximation properties of the model class.

Supplement G: Fixed-penalty simulation results (RMSE and wall-clock tables corresponding to Tables 1 and 2 under $\lambda = 10^{-5}$), including completion counts for capped cells.

Supplement H: A single-predictor base R reference implementation of the constrained monomial spline procedure.

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Appendix A: Derivation Using Lagrangian Multipliers

To derive the constrained penalized estimate, solve

$$\tilde{\beta} = \arg \min_{\beta} \left\{ \frac{1}{2} \|\mathbf{y} - \mathbf{X}\beta\|^2 + \frac{1}{2} \beta^\top \mathbf{\Lambda} \beta \right\} \quad \text{subject to} \quad \mathbf{A}^\top \beta = \mathbf{0}. \quad (19)$$

Introducing Lagrange multipliers δ and recalling $\mathbf{G} = (\mathbf{X}^\top \mathbf{X} + \mathbf{\Lambda})^{-1}$, the Lagrangian may be expressed as:

$$\mathcal{L} = -\mathbf{y}^\top \mathbf{X} \beta + \frac{1}{2} \beta^\top \mathbf{G}^{-1} \beta + \delta^\top \mathbf{A}^\top \beta. \quad (20)$$

Setting partial derivatives to zero:

$$\frac{\partial \mathcal{L}}{\partial \beta} = -\mathbf{X}^\top \mathbf{y} + \mathbf{G}^{-1} \beta + \mathbf{A} \delta = \mathbf{0}, \quad (21)$$

$$\frac{\partial \mathcal{L}}{\partial \delta} = \mathbf{A}^\top \beta = \mathbf{0}. \quad (22)$$

From (21), $\beta = \mathbf{G} \mathbf{X}^\top \mathbf{y} - \mathbf{G} \mathbf{A} \delta = \hat{\beta} - \mathbf{G} \mathbf{A} \delta$. Substituting into (22):

$$\mathbf{A}^\top \hat{\beta} - \mathbf{A}^\top \mathbf{G} \mathbf{A} \delta = \mathbf{0} \quad \implies \quad \delta = (\mathbf{A}^\top \mathbf{G} \mathbf{A})^{-1} \mathbf{A}^\top \hat{\beta}. \quad (23)$$

The constrained estimate is therefore

$$\tilde{\beta} = \hat{\beta} - \mathbf{G} \mathbf{A} (\mathbf{A}^\top \mathbf{G} \mathbf{A})^{-1} \mathbf{A}^\top \hat{\beta} = [\mathbf{I} - \mathbf{G} \mathbf{A} (\mathbf{A}^\top \mathbf{G} \mathbf{A})^{-1} \mathbf{A}^\top] \hat{\beta} = \mathbf{U} \hat{\beta}, \quad (24)$$

where $\mathbf{U}$ is the projection defined in (4). For non-zero constraints $\mathbf{A}^\top \beta = \mathbf{c} = \mathbf{A}^\top \beta_0$, the multiplier becomes $\delta = (\mathbf{A}^\top \mathbf{G} \mathbf{A})^{-1} \mathbf{A}^\top (\hat{\beta} - \beta_0)$ and the constrained estimate is $\tilde{\beta} = \mathbf{U} \hat{\beta} + (\mathbf{I} - \mathbf{U}) \beta_0$.

To verify the transformed OLS equivalence (Section 3.1), note that

$$\begin{aligned} \mathbf{U} \hat{\beta} &= (\mathbf{I} - \mathbf{G} \mathbf{A} (\mathbf{A}^\top \mathbf{G} \mathbf{A})^{-1} \mathbf{A}^\top) \hat{\beta} \\ &= \mathbf{G}^{1/2} (\mathbf{G}^{-1/2} \hat{\beta} - \mathbf{G}^{1/2} \mathbf{A} (\mathbf{A}^\top \mathbf{G}^{1/2} \mathbf{G}^{1/2} \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{G}^{1/2} \mathbf{G}^{-1/2} \hat{\beta}) \\ &= \mathbf{G}^{1/2} (\mathbf{y}^* - \mathbf{X}^* (\mathbf{X}^{*\top} \mathbf{X}^*)^{-1} \mathbf{X}^{*\top} \mathbf{y}^*) \\ &= \mathbf{G}^{1/2} \mathbf{r}^*, \end{aligned} \quad (25)$$

for $\mathbf{y}^* = \mathbf{G}^{-1/2} \hat{\beta}$ and $\mathbf{X}^* = \mathbf{G}^{1/2} \mathbf{A}$. The constrained solution is thus a left-multiplication of the OLS residual vector $\mathbf{r}^*$ by $\mathbf{G}^{1/2}$.

For non-zero constraints, set $\mathbf{y}^* = \mathbf{G}^{-1/2} \beta_0$ and keep $\mathbf{X}^*$ as above; then $\mathbf{G}^{1/2} \hat{\gamma}$ recovers the $(\mathbf{I} - \mathbf{U}) \beta_0$ component of $\tilde{\beta} = \mathbf{U} \hat{\beta} + (\mathbf{I} - \mathbf{U}) \beta_0$.

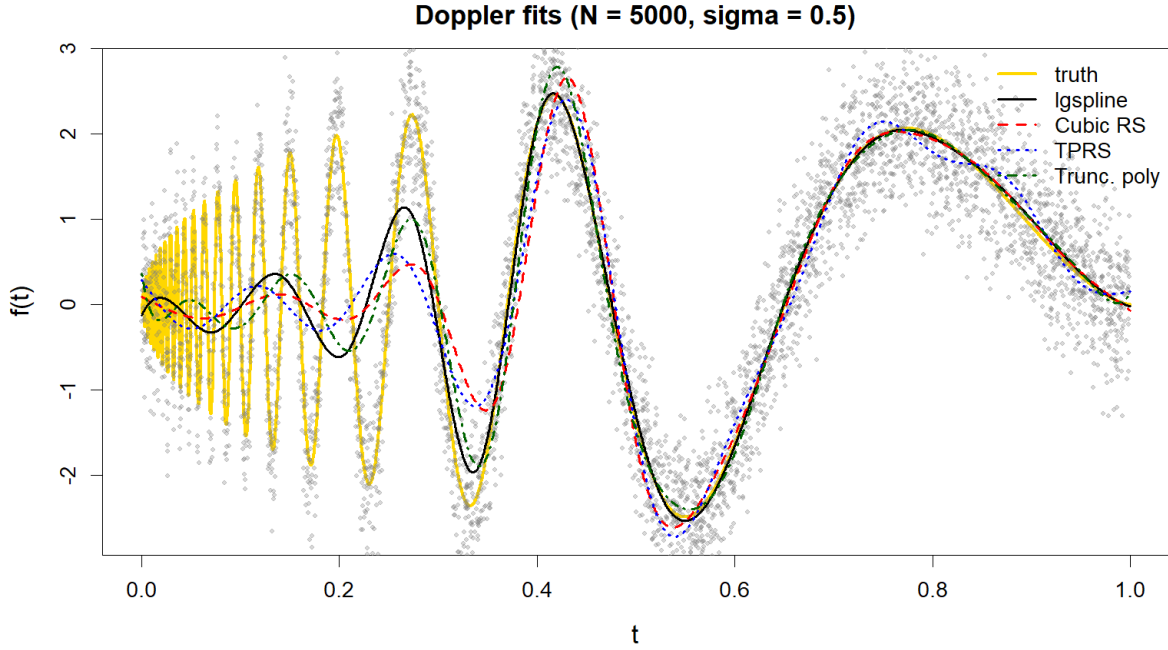


Figure 1: Representative fits on the Donoho-Johnstone Doppler function scaled by 5, at $N = 5,000$ and $\sigma = 0.5$ under automatic smoothness selection. Grey points show the noisy observations ($N = 5,000$); the gold solid line is the true function. The four fitted methods use matched basis dimensions ($K^* = 16$): `lgspline` (black solid), Cubic RS (red dashed), TPRS (blue dotted), and the truncated polynomial (dark green dash-dot). All four methods recover the low-frequency structure on $t \in [0.4, 1.0]$ closely, but none resolves the first high-frequency oscillation near $t = 0$ at this basis dimension; this is consistent with the tight clustering of RMSE values at $q = 1$ reported in Table 1.



Figure 2: Wall-clock time by sample size for each method, faceted by predictor dimension q and noise level σ . Solid lines show automatic smoothness selection; dashed lines show the fixed penalty. Points are means over $M = 24$ replicates. No wall-clock cap was imposed; every replication ran to completion. Exact numerical timings appear in Table 2.

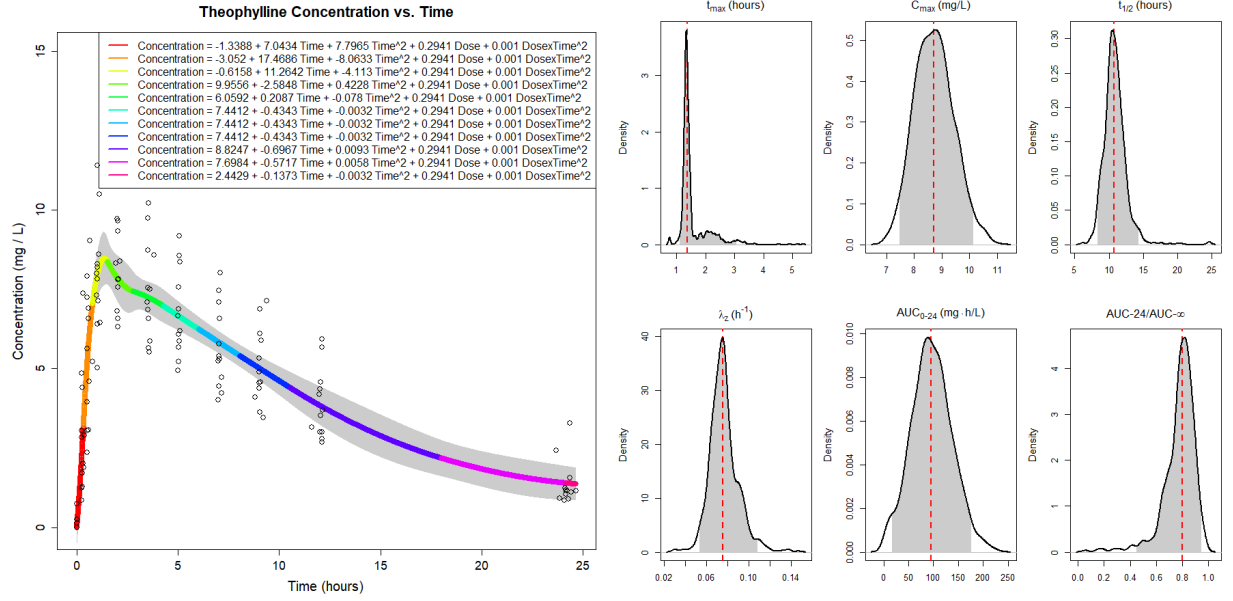


Figure 3: Fitted concentration-time curve (left) and posterior densities of the six pharmacokinetic summaries (right) for the Theophylline analysis. Shape constraints produce a monotone concave terminal-phase decay; the posterior densities carry the fully propagated uncertainty summarized in Table 4.

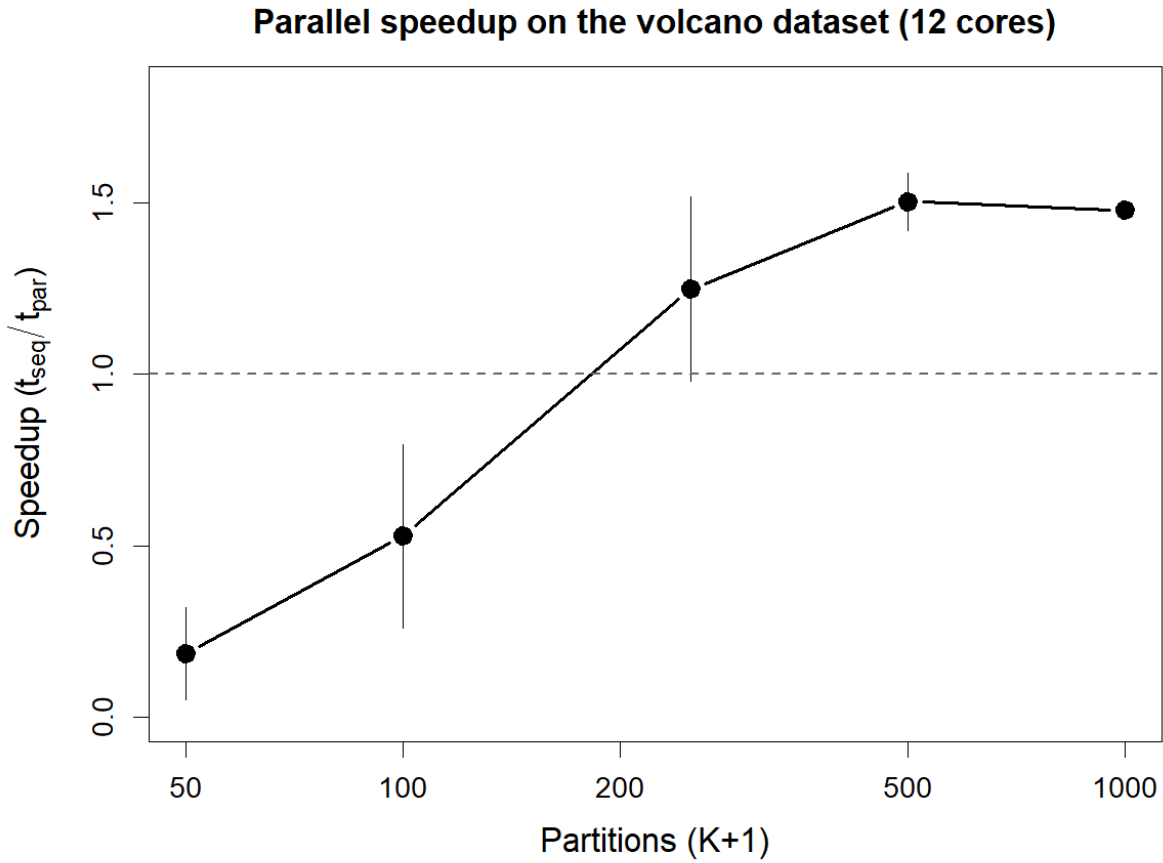


Figure 4: Parallel speedup of `lgspline` on the Maunga Whau volcano dataset as a function of partition count, with 12-core parallelism. Parallelism is a net cost below roughly $K + 1 = 200$, crosses break-even near $K + 1 = 250$, and plateaus at $\sim 1.5\times$ by $K + 1 = 500$. The plateau reflects the serial $P \times R$ constraint QR, which is not currently parallelized.

Volcano 3-D Map

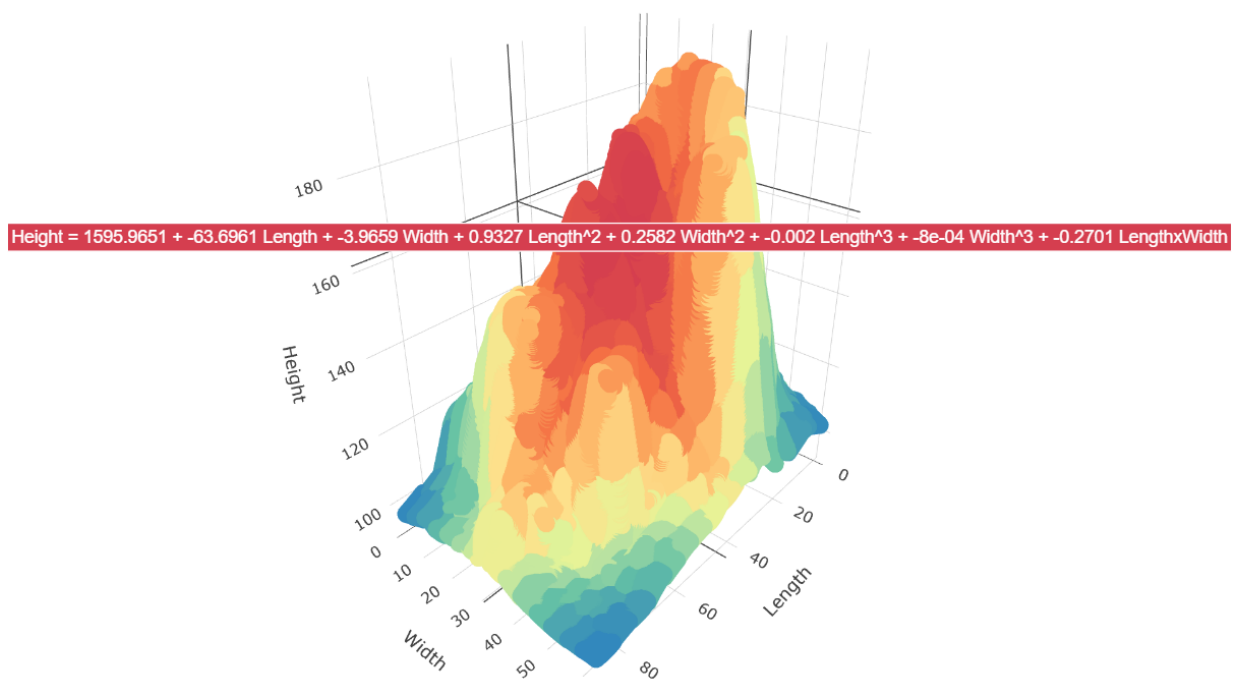


Figure 5: Illustrative `lgspline` fit to the Maunga Whau volcano dataset using a tuned model with $K + 1 = 250$ partitions (colored), fitted-value continuity only, pairwise linear interactions, and cubic expansions for each predictor (Width and Length). The displayed formula is the partition-specific polynomial equation used to generate the fitted surface, accessible by hovering over each partition with a cursor in the interactive `lgspline` output.